

# Ethical Considerations for Generative AI in Different Domains

Estimated time: 30 minutes

Welcome to the lab Ethical Considerations for Generative AI in Different Domains!

## Learning objectives

After completing this lab, you will be able to:

- Explore examples of using generative AI in different domains.
- Identify the ethical concerns regarding the usage of generative AI in different domains.

## Introduction

Professionals working in different domains leverage generative AI for several use cases. As they harness the benefits of generative AI, they also need to be aware of the limitations, concerns, and ethical implications associated with generative AI.

In this lab, you will experiment with generating content based on common use cases in different domains and analyze the response for correctness, authenticity, and other ethical concerns and considerations.

*Note: We have used the ChatGPT GPT-4.0 model for the exercise of this lab. Please note that the model's output may vary depending on different factors such as the phrasing of queries, model updates, etc.*

## Exercise 1: Evaluate generative AI's ability to generate output for the human resource (HR) domain

In this exercise, you will analyze the output generated by ChatGPT in response to the prompt for generating an email response to a job applicant informing them that their application was unsuccessful. We will analyze the output for any potential biases, inappropriate language, and fairness and empathy required for HR communication. To generate an output that is professional in content and tone, we need to ensure that the prompt includes relevant instructions.

### Step 1: Log in to ChatGPT

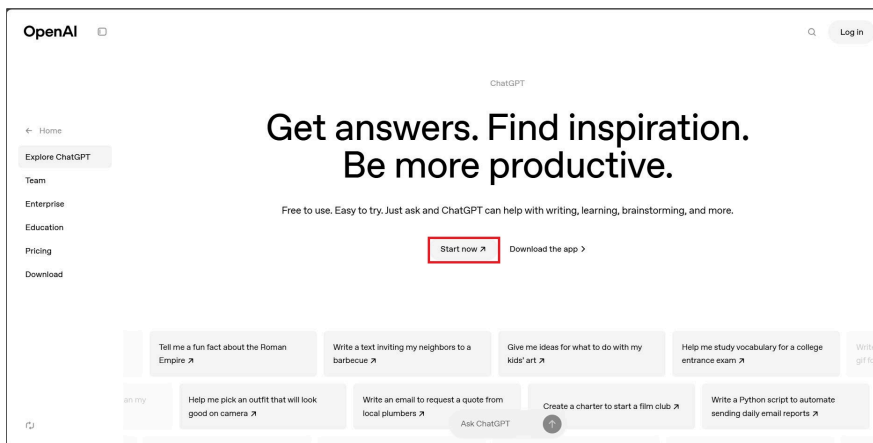
1. Launch [ChatGPT](https://chat.openai.com/) and sign in.

**Note:** If you cannot access ChatGPT by clicking on the link, copy the following URL and paste it in a browser.

<https://chat.openai.com/>

2. Once launched, you must sign up or login.

**Note:** You can create a free account on ChatGPT and use it for free.



a. To sign up, you will be required to enter your email ID and password. You will receive an automated verification email. After you click verification, you will need to enter your personal details and follow the steps to complete the sign up on your first login.

b. After sign up, you will reach the login page, as shown below.

# Welcome back

Email address\*

Continue

Don't have an account? [Sign Up](#)

OR

 Continue with phone

 Continue with Google

 Continue with Microsoft Account

 Continue with Apple

3. After login, you will view the OpenAI chatbot platform, ChatGPT, as shown below.

ChatGPT

User

What can I help with?

Message ChatGPT

Search

Reason

Create image

Code

Surprise me

Make a plan

Summarize text

More

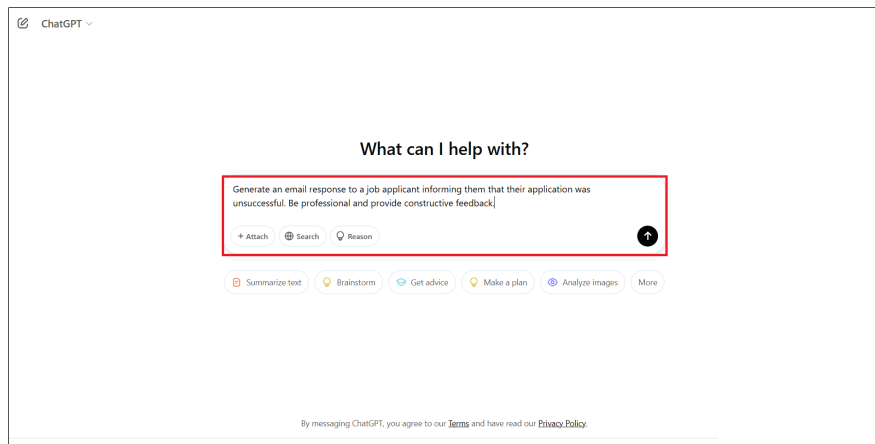
Get citation

ChatGPT can make mistakes. Check important info.

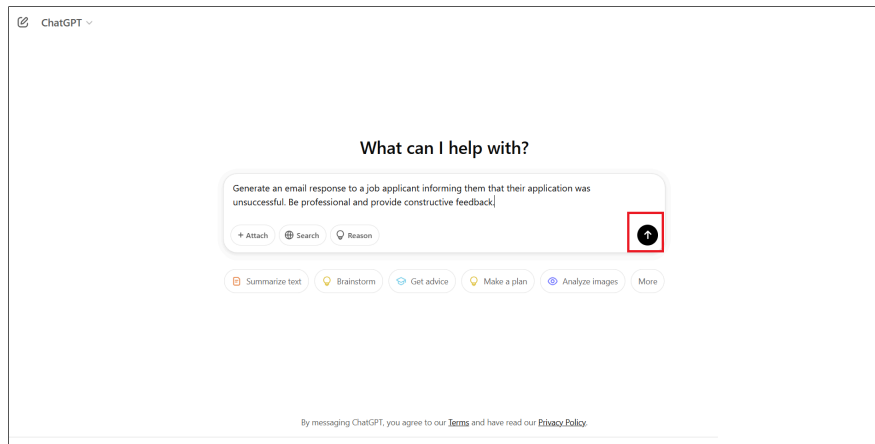
## Step 2: Generate the response

1. Let's enter the following prompt in the Message ChatGPT field.

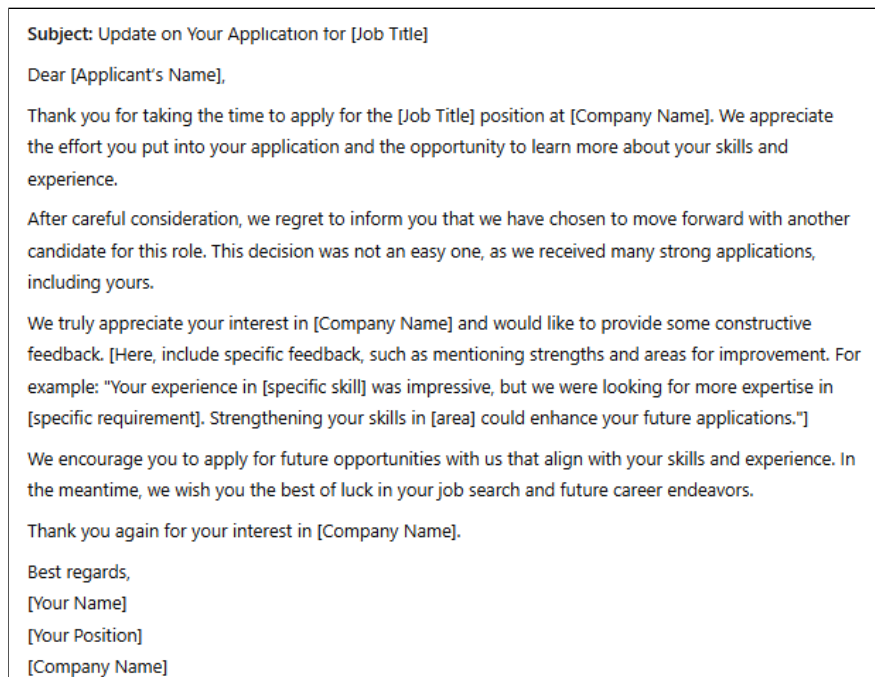
Generate an email response to a job applicant informing them that their application was unsuccessful. Be professional and provide constructive feedback.



2. Click the **Send Message** icon to generate the output.



3. We received the output shown in the following image.



If you analyze the content of the email, you will find that it begins by expressing gratitude for the applicant's interest and effort. The email maintains a professional and respectful tone throughout.

We also have to be considerate about biases. Fairness in HR communication is essential to mitigate biases. Every applicant should be treated with respect, and decisions should be based on merit. The email avoids using language that could be interpreted as biased or discriminatory.

In a more specific scenario, you also need to check the accuracy of the content, for example, the skills required for the role and shortcomings in the candidate's skills and experience to align with the role. The placeholder for constructive feedback suggests a personalized approach, which is beneficial for the applicant's growth.

On the whole, it is crucial to have a round of human review for context, bias, accuracy, and language.

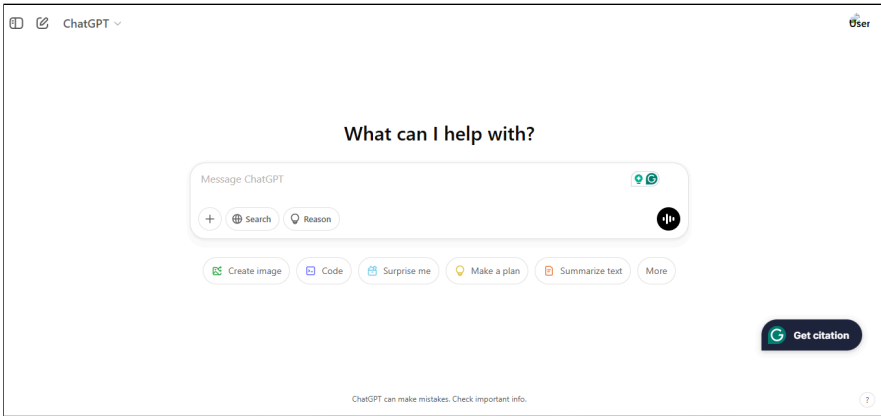
## Exercise 2: Use generative AI for content creation in the marketing domain

Professionals in the marketing domain can leverage generative AI to create engaging content for diverse use cases including:

- Create content for blog posts, articles, and social media content
- Draft emails for personalized communications based on user behavior and preferences
- Generate optimized meta descriptions and title tags for better search engine visibility
- Generate content for frequently asked questions and knowledge base articles
- Create personalized ad content based on user data and preferences
- Create surveys and questionnaires to gather valuable insights from the target audience
- Generate product descriptions for e-commerce websites

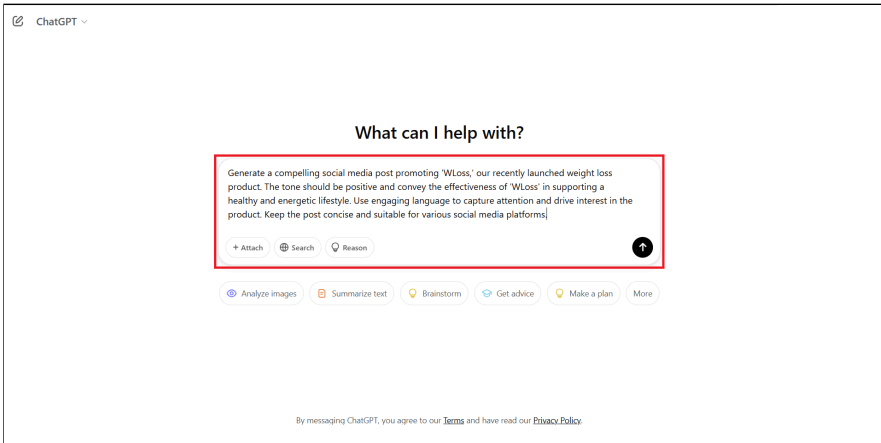
In this exercise, we will experiment with ChatGPT to create a social media post regarding a (fictitious) weight loss product. Furthermore, we will identify the ethical considerations and implications of using AI-generated content, such as media posts and blogs.

1. Ensure that you are logged in to the OpenAI chatbot platform, ChatGPT.

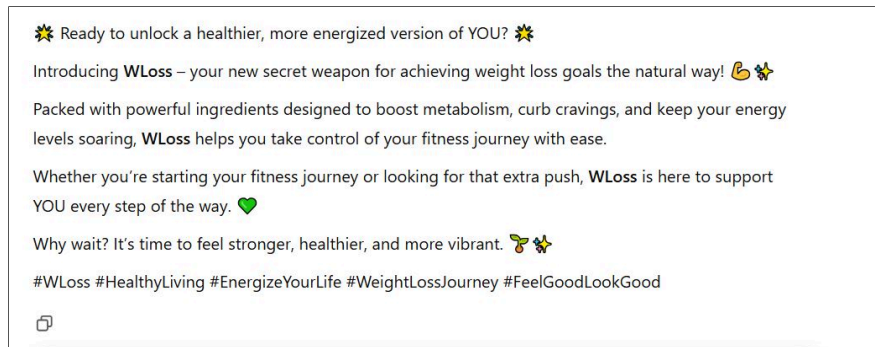


2. We will prompt ChatGPT to generate a social media post regarding a recently launched (fictitious) weight loss product. To produce an effective and relevant output, we should provide a prompt that includes suitable instructions regarding the different attributes, such as content, tone, and language. Let's enter the following prompt in the **Message ChatGPT** field.

Generate a compelling social media post promoting 'WLoss,' our recently launched weight loss product. The tone should be positive and convey the effe



3. Click the **Send Message** button to generate the output. We received the following output.



This social media post appears to be positive in tone and clear and engaging in language. In a real scenario, a post promoting a product should highlight the key features and benefits of the product.

Some important ethical considerations to keep in mind in case of social media post are:

- **Data privacy:** In such cases, when professionals use generative AI to create personalized content specific to an organization's products and services, they should avoid including any confidential data or information about the organization as part of the prompt.
- **Avoid biases:** Ensure that the content is free from biases.
- **Fact-based information:** For content intended to be used for promotional purposes, it is important to include evidence-based information and not any fact or information promoting unverified features or benefits. The responsibility of ensuring that AI-generated content aligns with factual information and does not propagate false narratives is a must.
- **Human review:** Implement a human review process to assess the content in terms of accuracy, context, biases, and other attributes.

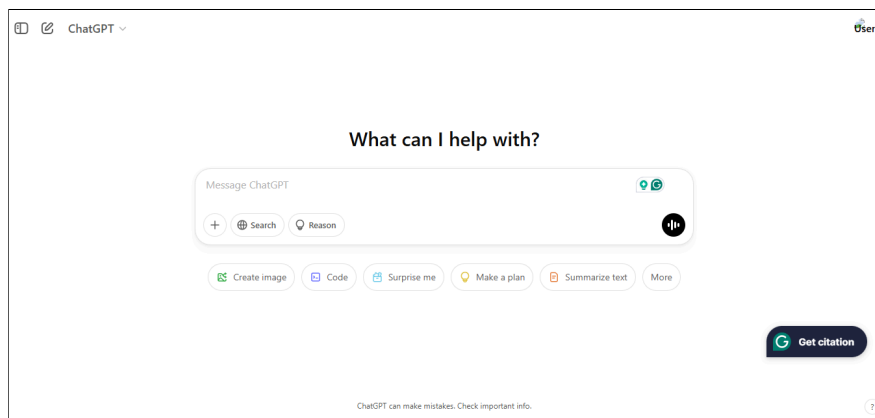
### Exercise 3: Use generative AI for data analysis based on customer data

Apart from content generation, another potential use case of generative AI in different domains, such as HR, sales, and marketing, is data analysis. In this exercise, we will understand how organizations and professionals can use generative AI to analyze and draw insights from marketing data and what are the ethical considerations and implications around this.

For the exercise, let's consider an example of a fictitious organization that has recently launched a new flagship smartphone. The marketing team is keen on understanding how customers perceive the product based on the reviews received from the customers. They considered using a generative model with natural language processing capabilities to help them analyze the feedback in terms of the recurring themes (such as camera, battery, and display) and the related sentiments (such as positive, mixed, and negative) about the different themes.

In the exercise, we will input reviews received from six (fictitious) customers in ChatGPT and provide a context-based prompt.

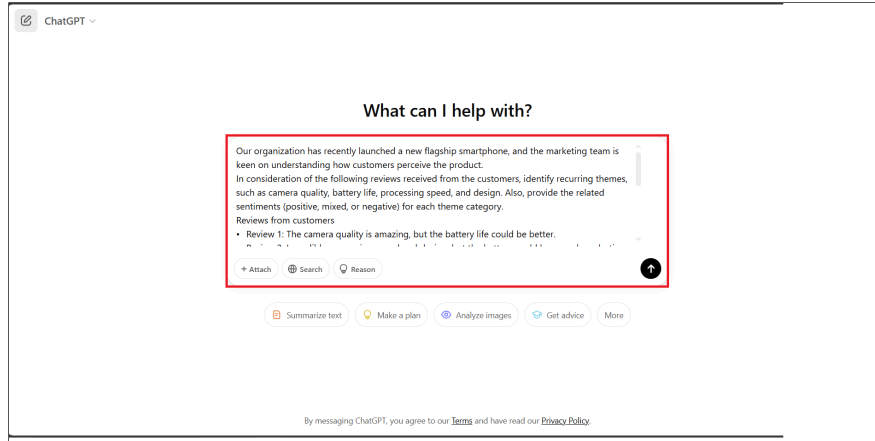
1. Ensure that you are logged in to the OpenAI chatbot platform, ChatGPT.



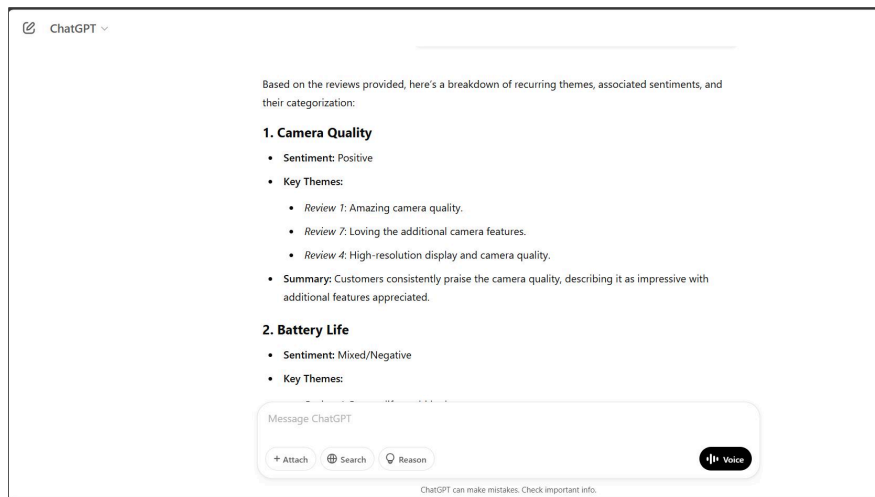
2. Let's enter the following prompt in the Message ChatGPT field. The prompt includes the content received from seven fictitious customers and instructions about the required analysis.

Our organization has recently launched a new flagship smartphone, and the marketing team is keen on understanding how customers perceive the product. In consideration of the following reviews received from the customers, identify recurring themes, such as camera quality, battery life, processing speed, and storage capacity. Provide a summary of the reviews from customers.

- Review 1: The camera quality is amazing, but the battery life could be better.
- Review 2: Incredible processing speed and design, but the battery could be more long-lasting.
- Review 3: The phone's display is brilliant, but the absence of expandable storage limits the overall storage capacity.
- Review 4: Like with the high-resolution display and camera quality, but the pre-installed apps are a bit too much. Would prefer a cleaner user interface.
- Review 5: The phone's sleek design is a head-turner. However, expandable storage should have been supported.
- Review 6: Overall, a solid performance, but the lack of expandable storage is a drawback for users with large media libraries.
- Review 7: The user interface is intuitive and easy to navigate. Loving the additional camera features.

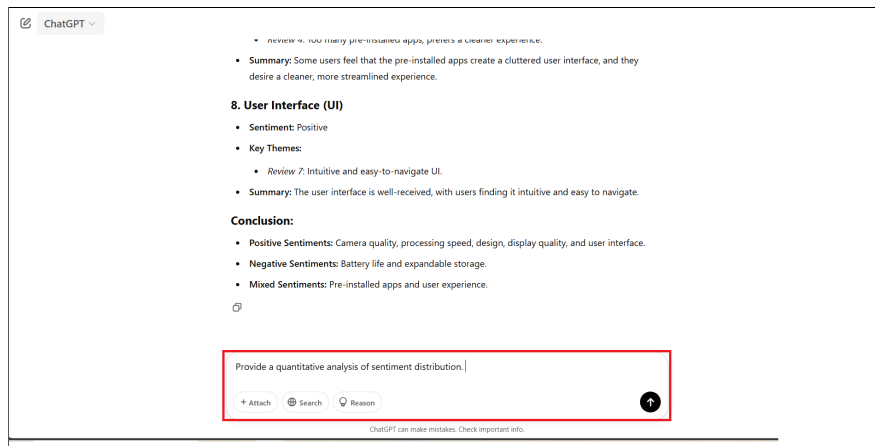


3. We received the following output from ChatGPT. You can notice that the response generated is organized and structured and provides a good analysis of the sentiment for different themes.

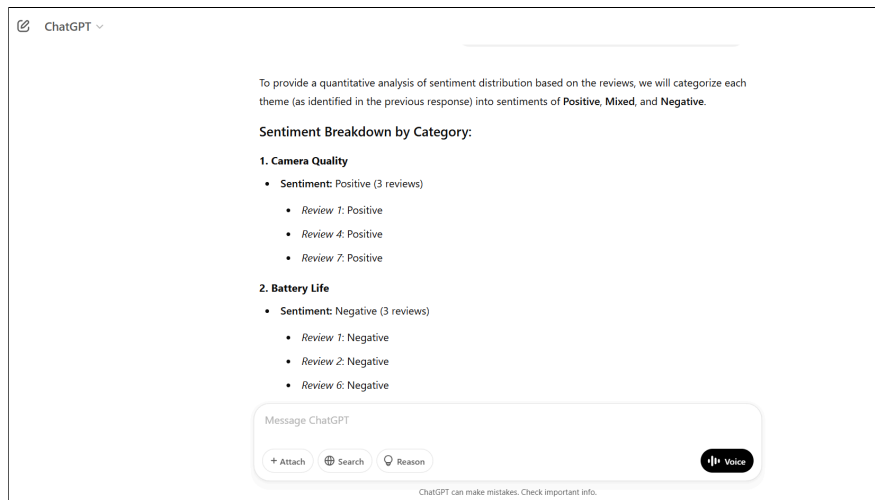


► Click here to see full output generated.

Provide a quantitative analysis of sentiment distribution.



5. The output generated by ChatGPT is shown in the image.



► [Click here to see the full output generated](#)

When marketing professionals use generative AI for analysis based on customer data, there are some important ethical considerations to consider.

- **Data privacy:** It is crucial to ensure that customer data is used and managed ethically and in compliance with privacy regulations. Professionals should respect customer consent and anonymize sensitive information to protect customer privacy.
- **Transparency:** Organizations and professionals need to be transparent about the use of generative AI in data analysis. They should communicate to the customers how AI is used, what data is analyzed, and what is the purpose of analysis.
- **Informed consent:** Organizations should inform customers that their data, including reviews and feedback, is being used for analysis. It is important to obtain explicit consent from the users for the use of their data in generative AI processes.
- **Bias:** Marketing professionals should be aware of and address any biases that could lead to unfair or discriminatory outcomes.

## Summary

Congratulations on completing the hands-on lab on Ethical considerations for generative AI in different domains.

In this lab, you experimented with a few use cases of generative AI in the HR and marketing domains. You identified the considerations and ethical implications around using generative AI for different use cases in these domains.

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